

Chapter 6 : Benzene and Its Derivatives

PSPM 2011 / 2012

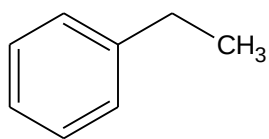
- Q. 1 (a) Compound **T** is formed from the reaction of benzene with chloroethane in the presence of AlCl_3 . Chlorination of **T** in light gives monochlorinated compounds **U** (major product) and **V**. Treatment of **T** in acidified solution of KMnO_4 produces **W**. Hydrolysis of **U** forms **X**; whereas reaction of **U** with nucleophile **Z** produces **Y**, $\text{C}_6\text{H}_5\text{CH}(\text{OCH}_2\text{CH}_3)\text{CH}_3$. Reagent **AA** transforms **X** to acetophenone while reagent **BB** converts the acetophenone to carboxylic acid.

Draw the structures of compounds **T**, **U**, **V**, **W** and **X** and state the reagents **AA** and **BB**. Name the reaction for the formation of **T** and write the mechanism. State whether the hydrolysis of **U** occurs via $\text{S}_{\text{N}}1$ or $\text{S}_{\text{N}}2$ mechanism and show the mechanism chosen. Give the nucleophile **Z**.

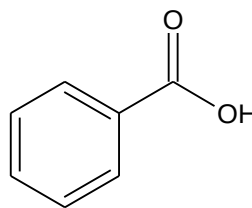
Compound **Y** is optically active. Draw a three-dimensional structural formula for an enantiomeric pair of **Y**.

Answer (a)

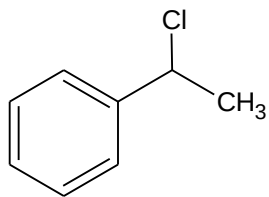
T =



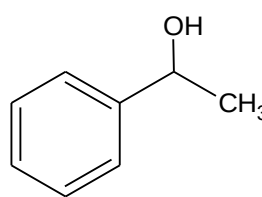
W =



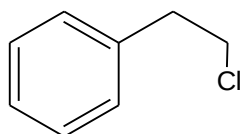
U =



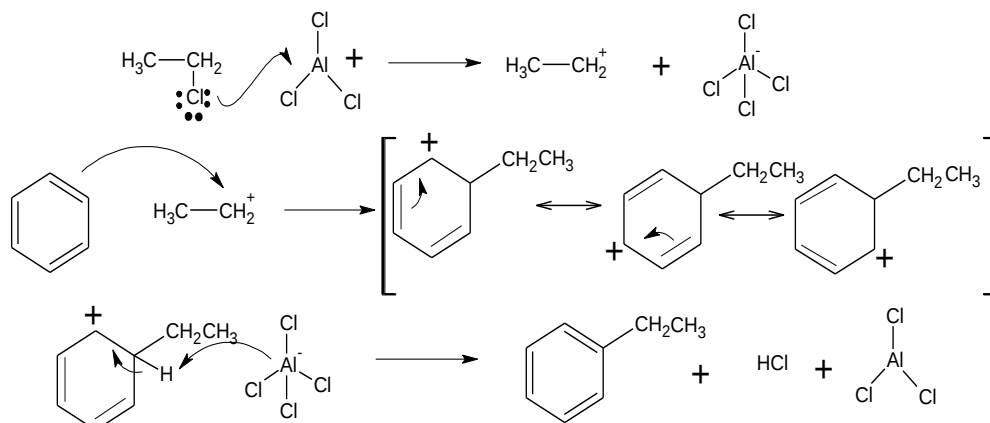
X =



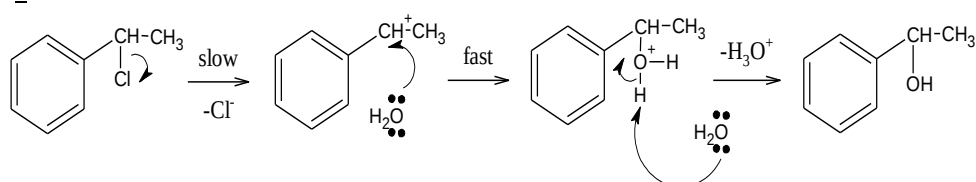
V =

AA = KMnO_4/H^+ or CrO_3/H^+ or $\text{Cr}_2\text{O}_7^{2-}/\text{H}^+$ BB = (i) I_2 , NaOH(ii) H_3O^+

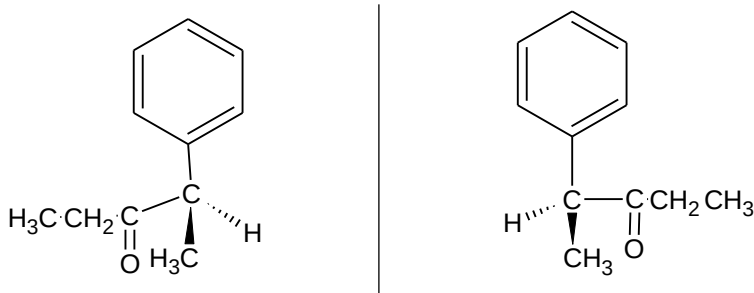
Friedel-Crafts alkylation



$\text{S}_{\text{N}}1$ mechanism



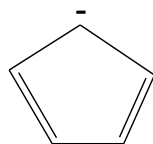
Z = $\text{CH}_3\text{CH}_2\text{OH}$



PSPM 2012 / 2013

- Q. 2 (a) Cyclopenta-1,3-diene anion shows aromaticity as in benzene. Explain the aromaticity of cyclopenta-1,3-diene anion. Draw two possible resonance structures for this compound.

9 m

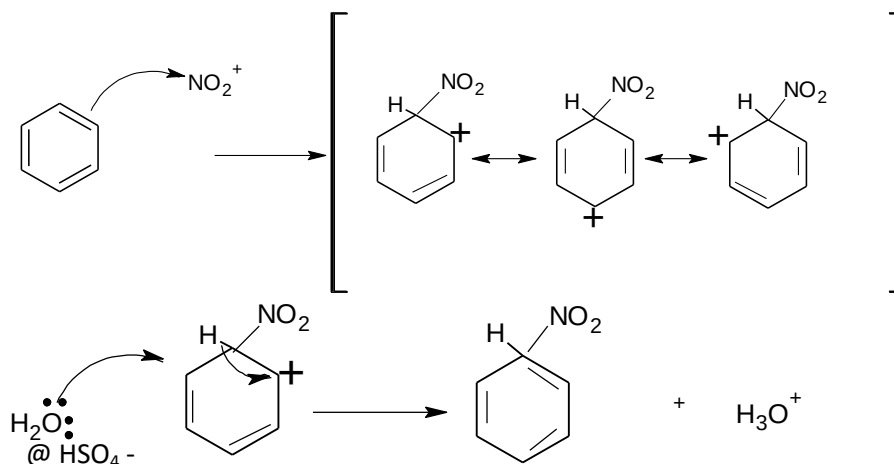
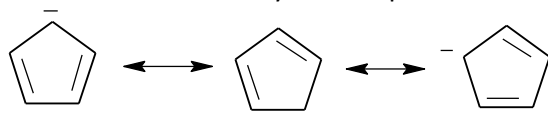


Cyclopenta-1,3-diene anion

Benzene will undergo electrophilic aromatic substitution with nitric acid, HNO_3 , in the presence of H_2SO_4 . Show the mechanism for this reaction.

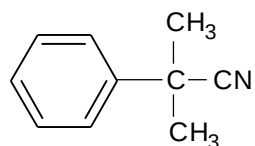
Answer (a) Cyclopenta-1,3-diene anion has **6 π electron** and follow Huckel's rule with **$n = 1$** @ **conjugated double bond**

All its **carbon atoms are sp^2 -hybridized / planar** and the **π electrons are delocalized** in the unhybridized p orbitals.



PSPM 2013 / 2014

Q. 3 (a) Explain the aromaticity of benzene. 10 m
Show how compound **W** can be prepared from benzene. Give all reagents and intermediates formed.



Compound **W**

Answer (a)

Benzene is aromatic because:

It obeys Huckel's rule

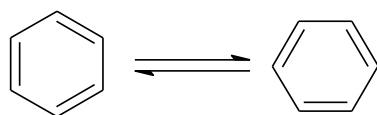
It is cyclic

It has conjugated C=C bonds

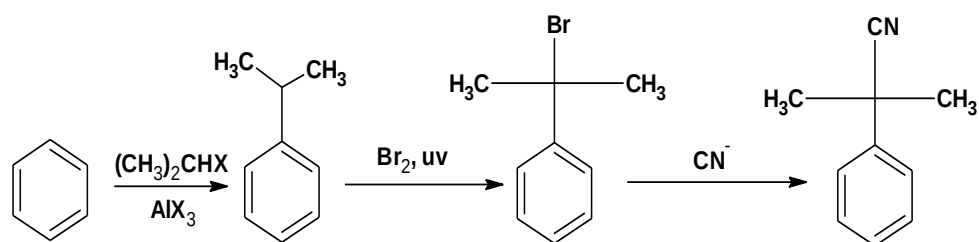
Pi electrons are delocalized within the ring

It is planar

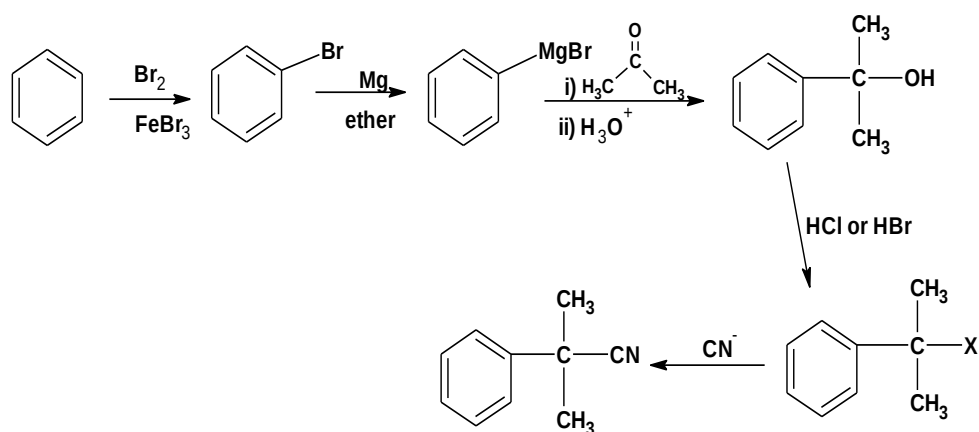
Kekule structures of benzene:



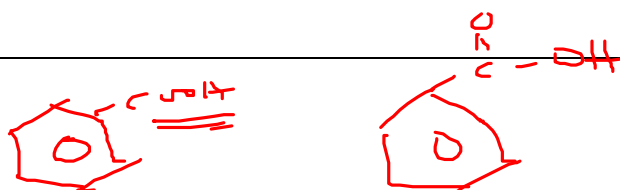
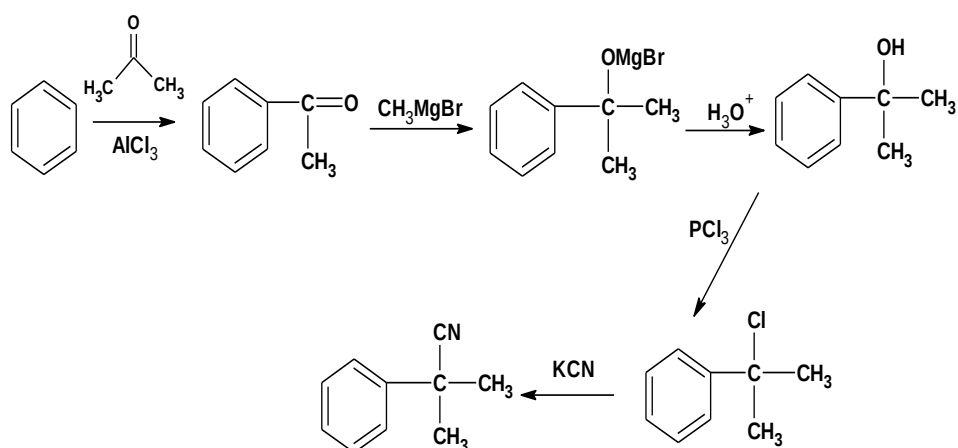
Preparation of **W**:



OR



OR

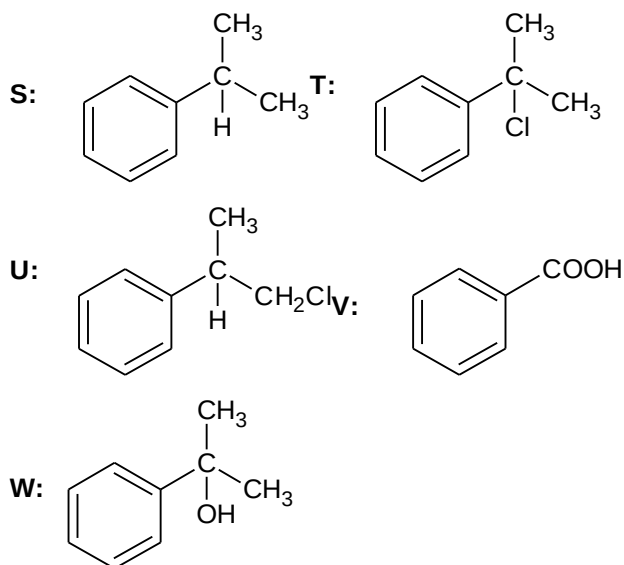


PSPM 2014 / 2015

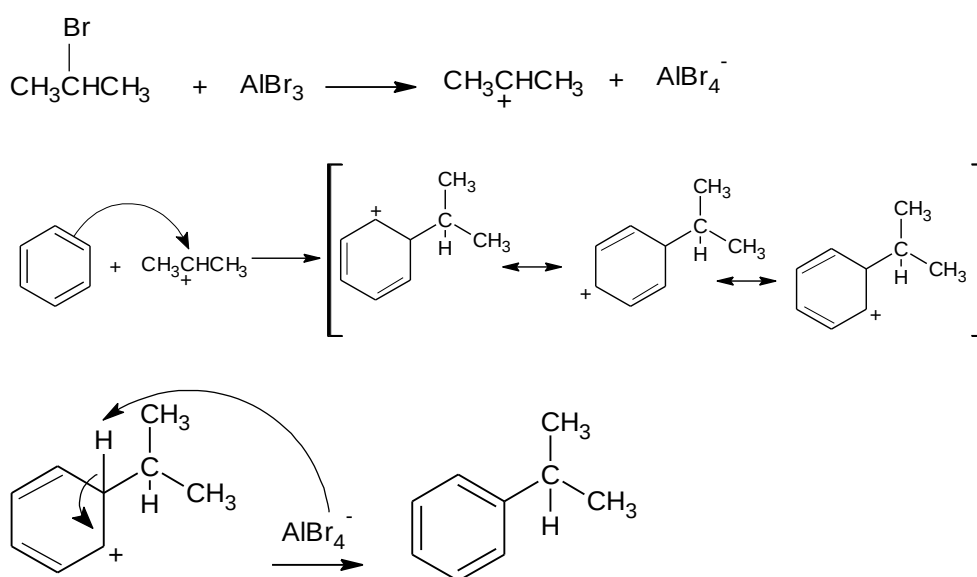
- Q. 4 (a) Compound **S** is formed when benzene reacts with 2-bromopropane in the presence of AlBr_3 . Chlorination of **S** in light gives two monochlorinated compounds, **T** (major product) and **U**. When **S** is treated with acidified solution of KMnO_4 , compound **V** is formed. Hydrolysis of **T** gives **W**. Draw structures of **S**, **T**, **U**, **V** and **W**. Name the reaction for the formation of **S** and write its mechanism.

10 m

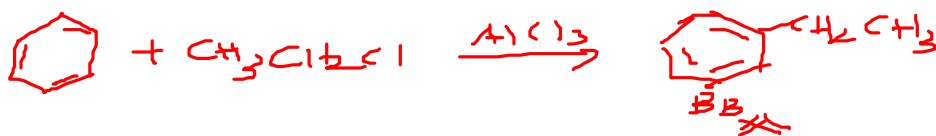
Answer



Name of reaction: Friedel-Crafts Alkylation

Mechanism:

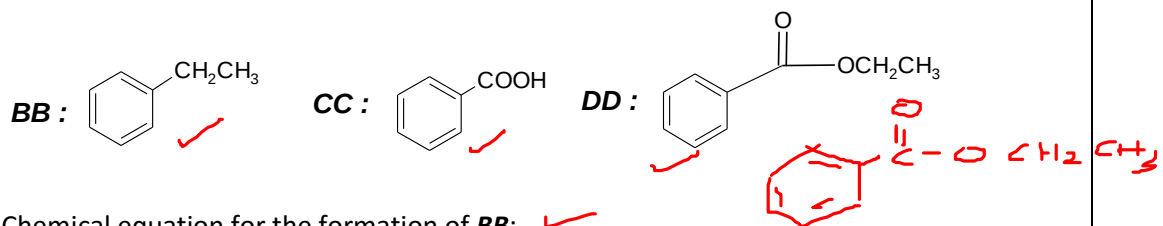
PSPM 2015 / 2016



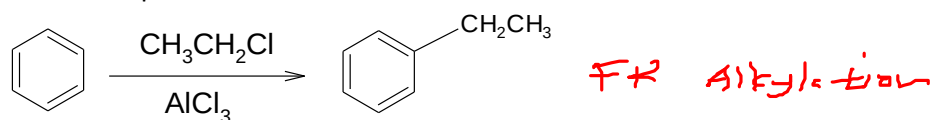
- Q. 5 (a) Compound BB is formed from the reaction of benzene with CH₃CH₂Cl in the presence of AlCl₃. Oxidation of BB with hot acidified solution of KMnO₄ gives CC. treatment of CC with ethanol in the presence of traces of sulphuric acid produces DD.

Name the type of reaction that converts benzene to BB. Suggest the structures of BB, CC and DD. Write the reaction equations for the formation of BB and DD. Write mechanism for the formation of BB.

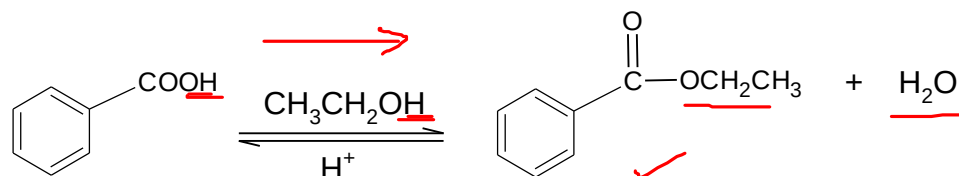
Answer (a) Electrophilic aromatic substitution ✓



Chemical equation for the formation of BB: ✓

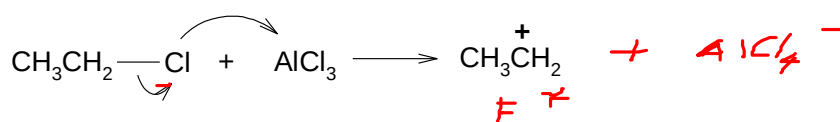


Chemical equation for the formation of DD: ✓

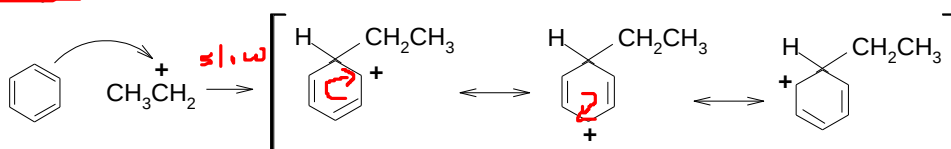


Mechanism:

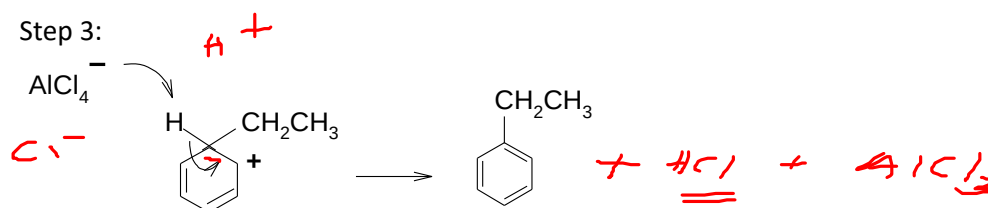
Step 1:



Step 2:



Step 3:

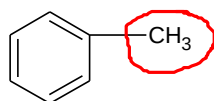
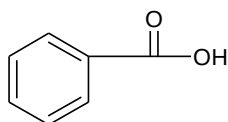
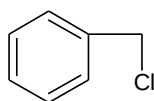
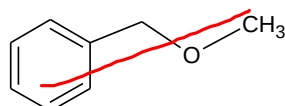


PSPM 2016 / 2017

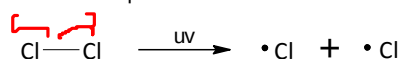
- Q. 6 (a) Compound **CC** is formed from the reaction of benzene with CH_3Cl in the presence of AlCl_3 . Oxidation of **CC** with reagent **DD** produce **EE**; chlorination of **CC** in the presence of light gives **FF**; whereas reaction of **FF** with methoxide ion, CH_3O^- , produces **GG**.

Suggest structural formulae of **CC**, **EE**, **FF**, **GG** and give reagent **DD**. Propose all the reactions involve in the initiation and termination steps for the chlorination of **CC** to **FF**. Name the reaction that convert to **CC** and write the mechanism involved.

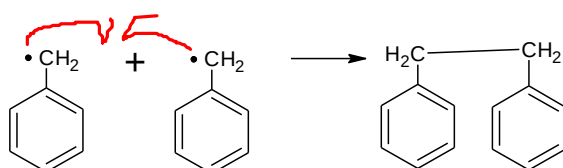
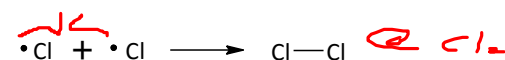
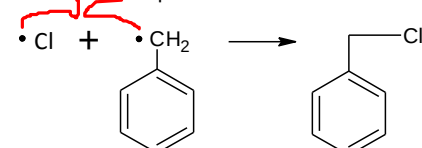
Answer (a)

CC

DD
 $\text{KMnO}_4, \text{H}^+, \text{heat}$ @ $\text{K}_2\text{Cr}_2\text{O}_7, \text{H}^+, \text{heat}$ @ i) $\text{KMnO}_4, \text{OH}^-, \text{heat}$
 ii) OH^-
EE

FF

GG


Initiation step

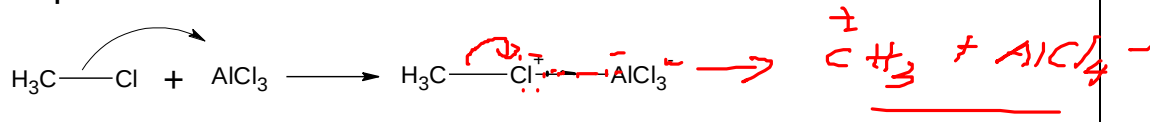


Termination step

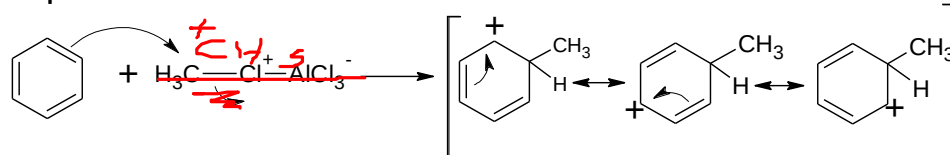


Name of reaction :Friedel-Crafts Alkylation

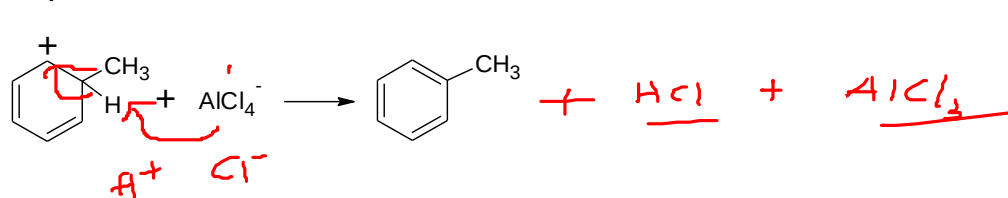
Step 1



Step 2



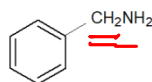
Step 3



PSPM 2017 / 2018



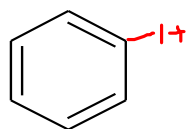
Q. 7 (a) Aromatic ring is less reactive towards chlorination than alkene. By using suitable equations, explain this statement. 10 m



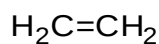
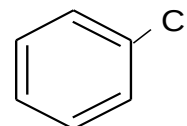
Benzylamine

Outline a synthetic pathway from benzene to benzylamine.

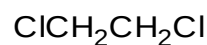
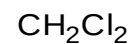
Answer (a)



+



+

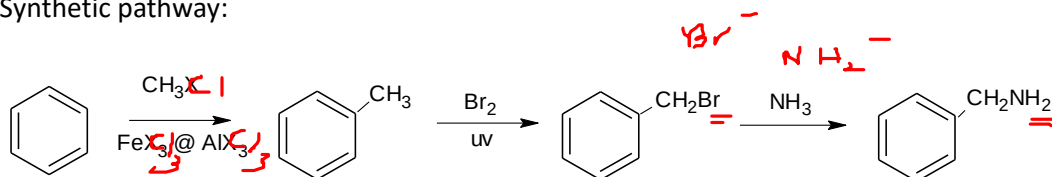


Benzene reacts with chlorine gas in the presence of Lewis acid to yield substituted product @ benzene undergoes electrophilic aromatic substitution. ✓

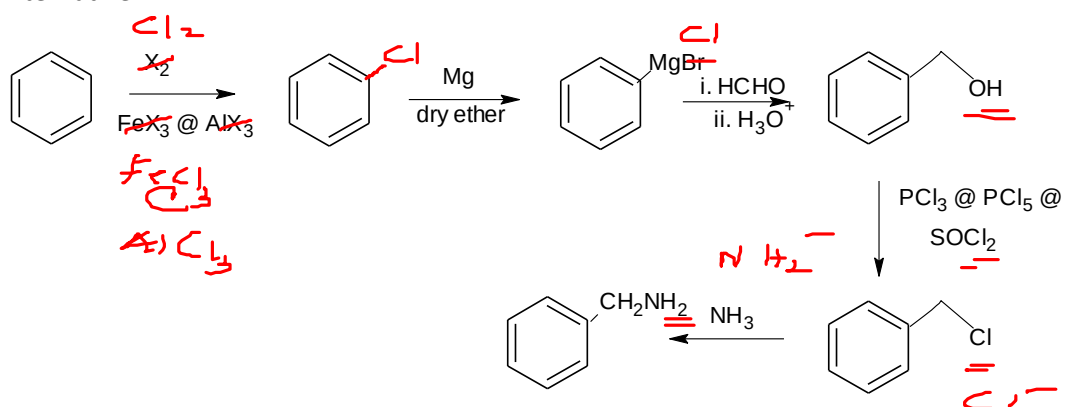
Alkene reacts with chlorine gas to yield addition product @ alkene undergoes electrophilic addition reaction. ✓

✓ Aromatic ring has delocalized electron which stabilizes the ring, thus less susceptible to electrophile attack.

Synthetic pathway:



Alternative:



*any acceptable answer